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## **AI Layoffs and Investor Reactions:**

### **Early Evidence from Public-Firm Layoff Announcements**

#### **1. Introduction**

A layoff is already an ambiguous signal for investors to read, since the same announcement can mean cost discipline if the firm is cutting unnecessary labor to protect margins, or it can mean weak demand, financial pressure, or management correcting earlier overhiring. AI complicates this further, because when a firm ties its layoff to AI adoption, the same labor cut can suddenly read as productivity-enhancing automation rather than ordinary restructuring. That reading can be fair, since replacing workers with technology may imply lower future costs and stronger cash flows, but the harder case is when the connection to AI is loose, vague, or mostly rhetorical, because if markets reward firms simply for attaching AI language to layoffs without much else changing, the framing itself starts to move prices in a way that has more to do with sentiment than with fundamentals.

I ask whether public firms that explicitly connect layoffs to AI adoption experience different stock-market reactions than firms announcing ordinary layoffs. The comparison holds the event type fixed, since every observation in the sample is a layoff event and the question is whether the AI framing changes the price response. I use a two-step event-study design where I first calculate cumulative abnormal returns around announcement dates, then regress those returns on a strict AI-layoff indicator with controls for layoff size, missing layoff-size data, and return source. I also look at shorter and longer windows to see whether any initial reaction lasts or fades, and at whether the response varies with layoff size, since a small AI-cited layoff may carry different information than a large one.

The behavioral angle of the paper has to do with how loaded AI information has become in the current market. Shiller's work on narrative economics is useful here because it treats economic narratives as forces that can move prices even when underlying fundamentals have not changed much, and AI-cited layoffs sit directly inside that tension because the same announcement can contain real productivity information and market-facing AI language at once. If investors initially overweight the AI framing, the reaction should appear at announcement and then weaken or reverse once the market has more time to figure out whether the layoff revealed actual productivity gain or mostly ordinary restructuring language.

Testing this requires building a sample that distinguishes strict AI-cited layoffs from layoffs that mention AI in passing, and that piece of work is harder than it sounds. There is no ready-made dataset of public firms tying layoffs to AI adoption, so I built one from public-firm layoff announcements and SEC filings, then manually coded whether each announcement makes a strict connection between the workforce reduction and AI. I start the sample in 2024 to focus on the post-ChatGPT disclosure period rather than the earlier pandemic hiring boom and its correction, which leaves some post-COVID restructuring in the background but reduces the chance that the results are mainly capturing the 2020 to 2022 overhiring cycle. Firms also have reasons to avoid AI-replacement language directly, since it is politically and reputationally sensitive, and Duolingo's 2025 AI-first memo illustrates why that language can create backlash. After deduping firm-date events and matching announcements to daily stock returns, the final sample contains 166 public-firm layoff events, of which 18 are strict AI-cited layoffs.

The results do not show that investors systematically reward firms for connecting layoffs to AI. Strict AI-cited layoffs do not produce statistically reliable abnormal returns relative to ordinary layoff announcements, the longer windows do not show a convincing reversal pattern, and the positive short-window layoff-size interaction is too imprecise to support a firm claim. The interpretation of my results is that AI framing alone does not appear to produce a reliable premium over ordinary layoff

announcements in the current public-firm sample, while the small treated group keeps the evidence early and leaves room for smaller or more uneven effects.

## **2. Literature and Theory**

Layoff announcements have never been a straightforward valuation signal, since investors can read the same labor cut as either cost discipline or as evidence of weakness, and Worrell, Davidson, and Sharma (1991) and Hallock (1998) document that announcement-window returns to layoffs have been mixed across earlier samples, with some events producing positive reactions when investors interpret the cut as restructuring or efficiency and others producing negative reactions when the cut signals weak demand or financial distress. That mixed pattern matters because it makes the expected sign of any layoff-related coefficient ambiguous from the beginning, and it implies that an AI-cited layoff cannot be treated as automatically positive just because it mentions AI, since it is still a layoff first with AI layered onto the underlying labor signal. The right control group is ordinary layoffs, since the goal is to isolate the effect of AI framing within the same corporate event.

An event-study design fits this question because the goal is to measure how investors react to a public announcement rather than to estimate what the long-run effect of the underlying corporate decision turns out to be, and MacKinlay (1997) provides the standard treatment of the method as it is used in corporate finance. The relevant event here is the layoff announcement, the treatment is whether that announcement explicitly connects the workforce reduction to AI adoption, and the cumulative abnormal return is the standard measure of how the market interprets the news during the days around the public release. This design asks whether the price response differs when the same labor cut is publicly framed as AI-related.

Overreaction to ambiguous or salient news is one of the most documented departures from rational pricing in behavioral finance, and De Bondt and Thaler (1985) describe the basic pattern in which investors place too much weight on recent or vivid information and then partially reverse as more information arrives. That logic motivates the paper's comparison between shorter and longer event windows, since rational repricing should generally persist while a reaction driven by salience or attention should partly weaken once the market has had

time to look more carefully at the disclosure. The paper does not require any particular reversal pattern from theory, but the window structure gives the behavioral interpretation a direct test.

Hirshleifer and Teoh (2003) argue that investors do not weigh every piece of disclosure equally and that the wording of an announcement can influence which part of the news investors treat as the most important signal, especially when the announcement is complex. In a layoff disclosure that also mentions AI, the AI language can guide attention toward a productivity-improvement reading and away from the underlying labor cut, even when the labor signal is the more economically material part of the announcement. The paper draws a strict line between disclosures that simply mention AI and disclosures that explicitly tie the workforce reduction to AI adoption because a general AI reference now appears in a large share of corporate disclosures, while the strict tie is the wording most likely to shift investor attention.

AI is one of the most loaded categories of corporate language at the moment, and Shiller's work on narrative economics treats economic narratives as forces that can move prices even when underlying fundamentals have not changed much. The implication for this paper is that the same wording can carry real information about future cost reductions while also making the announcement feel more forward-looking than the underlying cash-flow change would justify. An event study comparing strict AI-cited layoffs to ordinary layoffs cannot fully separate those two channels, but it can detect whether the average price response is different when the framing is present.

The first hypothesis is whether strict AI-cited layoffs produce different announcement-window cumulative abnormal returns than ordinary layoff announcements, with the sign left open since neither the layoff literature nor the behavioral literature predicts a confident direction. The second is whether any initial reaction persists or fades over longer windows, which would speak to whether the response reflects information absorption or attention-driven pressure. The third is whether the response varies with layoff size, since a larger workforce reduction makes the AI substitution claim more economically material while also raising the possibility that the firm is under more serious pressure that the AI framing happens to be covering. The contribution is to bring the layoff, event-study, attention, and narrative literatures together in a setting

prior work has not isolated directly, which is public-firm layoff announcements that explicitly tie the workforce reduction to AI adoption.

### **3. Data**

The sample is hand-built from public-firm layoff announcements, since no existing dataset captures the AI-cited layoff disclosure category needed for this paper. Every observation is already a layoff event, and the comparison is between announcements that explicitly tie the workforce reduction to AI and announcements that do not. The relevant date for each event is the date on which the layoff information became public, since the event study measures the market reaction to disclosed information instead of the internal date of the actual workforce decision.

The starting point is Layoffs.fyi, which records layoff announcements across a wide range of firms. Layoff entries are extracted from the tracker's public tabular display using OCR, since the records appear as rendered text on the page instead of as a downloadable file. The tracker on its own cannot serve as the sample because it includes private companies, subsidiaries, delisted firms, and entries with unclear ticker histories, so each event is matched to a public ticker when possible. Announcement text is then retrieved from press releases and SEC filings on EDGAR, since the manual AI coding requires the actual disclosure language and some AI-related workforce reductions are described mainly in formal filings.

Each announcement is coded into one of three categories by hand. A strict AI-cited layoff is one in which the announcement explicitly connects the workforce reduction to AI adoption, automation, AI-driven restructuring, or replacement of labor with AI-related technology. An ambiguous AI case contains AI language near the layoff or restructuring discussion but does not clearly identify AI as the reason for the workforce reduction. A non-AI case is an ordinary layoff announcement with no AI connection. The treatment variable is whether the announcement itself frames the labor cut as AI-related, which is a tighter definition than whether the firm uses AI or operates in an AI-related industry, since the wider version would now capture too much of the market. Manual coding is necessary because keyword searches alone overcount cases where AI and layoffs both appear in the same filing without being directly connected, and all coding decisions in this sample are made by a single coder against the criteria above.

The sample begins in 2024 to focus on the post-ChatGPT AI disclosure period and to reduce the influence of the earlier pandemic hiring boom and its immediate correction. The cutoff still leaves some post-COVID restructuring in the background, since some firms may continue to correct earlier overhiring, though it limits how much of the resulting variation could come from the 2020 to 2022 labor-cycle reversal. After coding, I deduplicate events by firm-date so the same layoff is counted only once when it appears in both Layoffs.fyi and EDGAR, and each remaining event is matched to a public ticker so the announcement can be linked to daily stock returns.

Daily return data come from CRSP62 where available, with Yahoo Finance filling in observations outside CRSP62 coverage, and the regressions in Section 4 include controls for the resulting hybrid return source. Jamf is excluded because usable daily return data could not be recovered. 2U/TWOUQ is excluded because its event-window returns are dominated by severe financial distress and bankruptcy-style trading, which makes the price reaction difficult to read as a normal response to AI-layoff information. The sensitivity to this exclusion is reported later in the paper instead of being absorbed silently into the sample.

Table 1 summarizes the final sample. The main analysis sample contains 166 public-firm layoff events, of which 18 are strict AI-cited layoffs, and the subsample with usable layoff size contains 160 events including 15 strict AI events. The treated group is small, which limits statistical power, though the size of the group is also part of the setting, since the sample measures publicly disclosed AI-layoff framing instead of the full set of layoffs that may have been caused by AI internally. Some firms may use AI to reduce labor without saying so directly, while others may mention AI too vaguely for the announcement to meet the strict coding bar.

**Table 1. Sample Construction**

| Step                                                               | Events | Strict AI events |
|--------------------------------------------------------------------|--------|------------------|
| Hybrid CRSP/Yahoo event sample before distressed-outlier exclusion | 167    | 19               |
| Exclude 2U/TWOUQ distressed outlier                                | -1     | -1               |
| Final main analysis sample                                         | 166    | 18               |
| Known layoff-size subsample                                        | 160    | 15               |

**Notes:** This table summarizes the construction of the final event-study sample. The sample begins with a hybrid CRSP/Yahoo public-firm layoff event sample. 2U/TWOUQ is excluded from the main analysis because its event-window returns appear dominated by severe financial distress and bankruptcy-style trading, making them difficult

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to interpret as a normal response to AI-layoff information. The final main sample contains 166 public-firm layoff events, including 18 strict AI-cited layoffs. The known layoff-size subsample contains 160 events, including 15 strict AI-cited layoffs.

#### 4. Methodology

In the event study, the unit of observation is a firm-layoff announcement, and the event date is the date on which the layoff information became public, matching the disclosure date used in the Data section. Daily abnormal returns are computed as the firm's return on day  $t$  minus the return on the market proxy on the same day.

$$AR_{it} = R_{it} - R_{mt}$$

In this equation,  $AR_{it}$  is the abnormal return for firm  $i$  on trading day  $t$ ,  $R_{it}$  is the firm's daily return, and  $R_{mt}$  is the market return. The market-adjusted approach fits the question because the paper is measuring investor reaction to public announcements, and it avoids relying on a long pre-event estimation window that could be affected by earlier firm disclosures.

Cumulative abnormal returns are computed as the sum of daily abnormal returns over fixed windows around the event date.

$$CAR_{it}[a,b] = \sum_{t=a}^b AR_{it}$$

The paper uses four windows.  $CAR[-1,+1]$  captures the immediate market reaction around the announcement,  $CAR[-1,+5]$  and  $CAR[-1,+10]$  extend the window to allow for short-run reassessment, and  $CAR[+3,+10]$  focuses on post-announcement movement after the first few trading days have passed.

The main regression estimates whether strict AI-cited layoff announcements have different cumulative abnormal returns from the comparison group of layoff announcements that do not make that connection.

$$CAR_{it} = \alpha + \beta_1 \text{StrictAI}_{it} + \gamma \text{Controls}_{it} + \varepsilon_{it}$$

$\text{StrictAI}_{it}$  equals one if the announcement explicitly connects the workforce reduction to AI adoption and zero otherwise. Ambiguous AI cases are pooled with non-AI announcements in the main specification because the test is centered on whether explicit AI framing produces a different price response. The controls include the natural log of layoff size, an indicator for events where layoff size could not be recovered from the

announcement, and a return-source fixed effect for whether returns come from CRSP62 or Yahoo Finance.

Table 3 reports the regression both without controls and with this control set, so the first specification shows the raw difference associated with strict AI framing while the controlled specification asks whether any difference remains after accounting for layoff size and return source.

The H3 specification adds an interaction between the strict AI indicator and centered log layoff size.

$$CAR_i = \alpha + \beta_1 \text{StrictAI}_i + \beta_2 \text{LogLayoffSize}_i + \beta_3 (\text{StrictAI}_i \times \text{LogLayoffSize}_i) + \gamma \text{Controls}_i + \varepsilon_i$$

Here  $\beta_3$  measures whether the AI-layoff effect varies with the size of the workforce reduction. Centering log layoff size makes  $\beta_1$  refer to the strict AI difference at the average layoff size in the usable-size sample, while  $\beta_3$  captures how that difference changes as layoff size moves above or below that average. The sign is left open because a larger AI-cited layoff could be read as a more economically material productivity claim or as evidence of deeper pressure that the AI framing happens to be covering. This interaction sample is restricted to events where layoff size is observed, leaving 160 events with 15 strict AI cases.

Inference uses HC1-robust p-values as the baseline, with industry-month clustered p-values and wild cluster bootstrap p-values reported as small-sample checks. The clustered p-values allow events in the same industry and month to be correlated, while the wild cluster bootstrap is useful because the strict AI group is small and standard clustered inference can be fragile in that setting.

## 5. Results

Figure 1 plots the cumulative average abnormal return path by announcement type around the event date, using the Yahoo-available daily return panel. The figure is descriptive rather than a formal test, and the plotting sample is slightly smaller than the 166-event regression sample because not every observation has a complete Yahoo daily return series. The strict AI and non-AI series move closely with each other across the window, with no sustained separation between them, which suggests that any AI-layoff premium in this sample is small enough that it is not visible in the raw averaged return path.

Table 3 reports the main CAR regressions across the four event windows, both without controls and with controls for layoff size, missing layoff-size data, and return source. In the bare specification, the immediate-window coefficient on Strict AI for  $CAR[-1,+1]$  is -0.0021 with HC1  $p = 0.938$ , which is



statistically and economically close to zero. Adding controls moves the immediate coefficient to 0.0144 with HC1  $p = 0.597$ , which is also statistically insignificant. The longer windows follow the same pattern. The strict AI coefficient is positive in the  $[-1,+5]$  and  $[-1,+10]$  windows, but the  $p$ -values remain well above conventional thresholds, and the  $[+3,+10]$  reversal window is also statistically insignificant. Taken as a test of H1, these regressions do not show a statistically reliable difference between strict AI-cited layoffs and ordinary layoff announcements.

The same results also weaken the overreaction interpretation in H2. If investors initially overweighted AI framing, the immediate announcement window should show a reaction large enough to fade or reverse later. The immediate coefficient is already close to zero, and the post-announcement window does not show a reliable correction, with  $CAR[+3,+10]$  of 0.0166 in the bare specification and -0.0062 in the controls specification. This does not prove that investors process AI disclosures perfectly, but it means the data do not show the temporary price pressure that the overreaction prediction would require.

Table 4 reports the layoff-size interaction regressions for H3, using the 160-event subsample with usable layoff-size data and 15 strict AI cases. The interaction between Strict AI and centered log layoff size is positive in the shorter windows, with coefficients of 0.0305 for  $CAR[-1,+1]$  and 0.0378 for  $CAR[-1,+5]$ . This is consistent with the possibility that AI framing may matter more when attached to a larger workforce reduction, but the estimates are too imprecise to support a firm claim. The wild cluster  $p$ -values are 0.531 and 0.557 in those two windows, and the longer-window interactions weaken further. The size pattern is directionally interesting, but it is not statistically reliable evidence that larger AI-cited layoffs receive more favorable market reactions.

The 2U/TWOUQ sensitivity check in Table 5 mainly illustrates how fragile inference can be when strict AI-cited announcements are still rare. Including 2U pulls the immediate-window coefficient to -0.0468 with  $p = 0.358$ , while excluding it leaves the final-sample coefficient at -0.0021 with  $p = 0.938$ . Since 2U's return path is dominated by severe financial distress and bankruptcy-style trading, the final sample excludes it for economic reasons, and the sensitivity check reinforces why these estimates should be read as early evidence.

The results point in a consistent direction across the figure, the main regressions, and the interaction tests.

The current public-firm sample does not show a statistically reliable AI-layoff premium, does not show a convincing reversal pattern, and does not provide statistically reliable evidence that layoff size changes the AI-framing response. The estimates are too imprecise to rule out smaller or uneven effects, but the evidence does not support the prediction that investors consistently reward firms for attaching AI language to labor cuts.

**Table 3. Main CAR Regressions**

| Window      | Spec     | N   | Strict AI coefficient | HC1 SE | HC1 p-value | Cluster p-value | Wild cluster p-value |
|-------------|----------|-----|-----------------------|--------|-------------|-----------------|----------------------|
| CAR[-1,+1]  | Bare     | 166 | -0.0021               | 0.0273 | 0.938       | 0.94            | 0.953                |
| CAR[-1,+1]  | Controls | 166 | 0.0144                | 0.0272 | 0.597       | 0.662           | 0.73                 |
| CAR[-1,+5]  | Bare     | 166 | 0.0363                | 0.0317 | 0.251       | 0.304           | 0.498                |
| CAR[-1,+5]  | Controls | 166 | 0.0432                | 0.033  | 0.191       | 0.271           | 0.502                |
| CAR[-1,+10] | Bare     | 166 | 0.0283                | 0.035  | 0.418       | 0.461           | 0.562                |
| CAR[-1,+10] | Controls | 166 | 0.0181                | 0.0368 | 0.623       | 0.671           | 0.697                |
| CAR[+3,+10] | Bare     | 166 | 0.0166                | 0.0227 | 0.463       | 0.251           | 0.491                |
| CAR[+3,+10] | Controls | 166 | -0.0062               | 0.0234 | 0.792       | 0.733           | 0.775                |

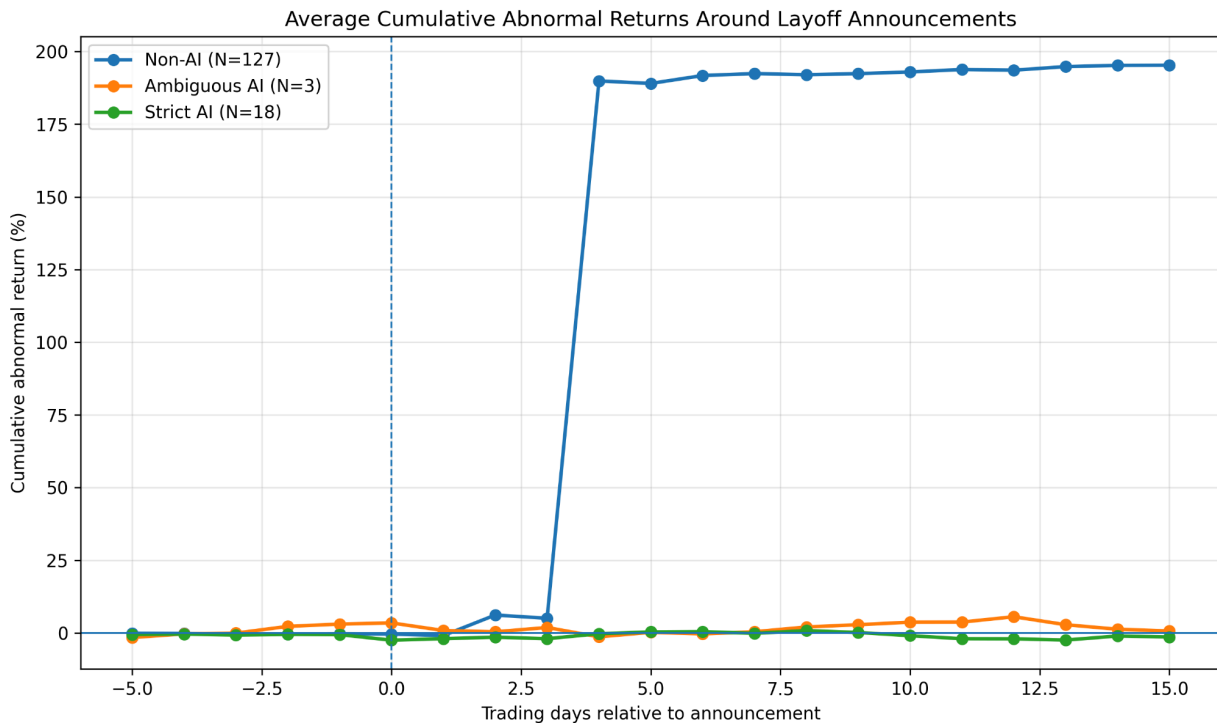
**Notes:** The dependent variable is cumulative abnormal return over the listed event window. The coefficient reported is the coefficient on Strict AI Layoff, which equals one when the layoff announcement explicitly connects the workforce reduction to AI adoption and zero otherwise. Bare specifications include only the Strict AI indicator. Control specifications include log layoff size, a missing layoff-size indicator, and return-source fixed effects. HC1 p-values use heteroskedasticity-robust standard errors. Cluster p-values cluster by industry-month. Wild cluster p-values are reported as a small-sample robustness check.

**Table 4. Layoff-Size Interaction Regressions**

| Window          | N   | Strict AI events | Strict AI × Log Layoff Size | HC1 SE | HC1 p-value | Cluster p-value | Wild cluster p-value |
|-----------------|-----|------------------|-----------------------------|--------|-------------|-----------------|----------------------|
| CAR[-1,+1]      | 160 | 15               | 0.0305                      | 0.0206 | 0.138       | 0.149           | 0.531                |
| CAR[-1,+5]      | 160 | 15               | 0.0378                      | 0.0285 | 0.184       | 0.181           | 0.557                |
| CAR[-1,+1<br>0] | 160 | 15               | 0.0214                      | 0.0336 | 0.524       | 0.518           | 0.654                |
| CAR[+3,+1<br>0] | 160 | 15               | -0.0024                     | 0.0216 | 0.912       | 0.906           | 0.909                |

**Notes:** The dependent variable is cumulative abnormal return over the listed event window. The coefficient reported is the interaction between Strict AI Layoff and centered log layoff size. The sample is restricted to events with usable layoff-size data. Positive coefficients suggest that larger AI-cited layoffs are associated with more favorable CARs relative to ordinary layoffs, but the estimates are not statistically significant. HC1 p-values use heteroskedasticity-robust standard errors, cluster p-values cluster by industry-month, and wild cluster p-values are reported as a small-sample robustness check.

**Plot 1**



**Notes:** This figure plots cumulative average abnormal returns by announcement type around the layoff announcement date. Strict AI events explicitly connect the layoff to AI adoption. Ambiguous AI events contain AI-related language but do not clearly identify AI as the cause of the layoff. The figure is descriptive and uses the Yahoo-available daily return panel, so the plotting sample is slightly smaller than the final 166-event regression sample.

## 6. Discussion

The results point toward a cautious interpretation of AI-cited layoffs in the current public-firm sample, since strict AI-cited layoffs do not earn a statistically reliable premium over ordinary layoff announcements even though the market has had strong reasons over the last few years to associate AI language with productivity gains, margin expansion, and future cost savings. If investors were rewarding AI framing in a consistent way, the strict AI group should have separated more clearly from ordinary layoffs in the immediate window, and

some of that difference should have appeared across the longer windows as well. Instead, the immediate coefficient sits close to zero after excluding 2U/TWOUQ, the longer windows remain imprecise, and the post-announcement window does not show a reliable reversal. That pattern suggests that investors are still treating AI-cited layoffs as ambiguous labor-cut announcements, rather than letting the AI label turn the announcement into favorable news on its own.

This interpretation makes more sense once the ambiguity of layoffs is taken seriously, because a workforce reduction can still carry negative information even when the firm describes it through AI adoption, automation, or restructuring language. Investors may see potential labor-cost savings, but they may also see weak demand, operating pressure, or management correcting earlier hiring mistakes. AI framing can make the announcement sound more forward-looking, especially when the firm presents the cut as part of a shift toward technology, while the market still has to decide whether the announcement reflects real productivity improvement or ordinary cost cutting with more attractive language attached. This also weakens the overreaction interpretation, since the immediate AI coefficient is already close to zero and leaves little room for the kind of later correction that a temporary mispricing argument would predict.

The layoff-size interaction is the most interesting secondary result, although it remains weak and imprecise. The interaction between Strict AI and log layoff size is positive in the shorter windows, which fits the possibility that AI framing may matter more when the workforce reduction is large enough to look economically material. A small AI-cited cut can sound like vague positioning, while a larger AI-cited cut may look more like a real shift from labor toward technology. The problem is that the interaction sample contains only 15 strict AI events with usable layoff size, and the wild cluster p-values are far from conventional levels. The size pattern is worth mentioning as a direction for later work, but it should not be treated as evidence that markets reward larger AI-cited layoffs in the current sample.

The small treated group also says something about the disclosure setting itself, since only 18 of the 166 public-firm layoff announcements in the final sample explicitly connect the workforce reduction to AI adoption under the strict coding rule. That scarcity is informative because the paper measures disclosed

AI-layoff framing, not every layoff that may have been internally influenced by AI. Firms may be using AI to reduce labor without saying so directly, especially when replacement language can create reputational or political backlash, and other firms may mention AI in the same filing while still framing the actual layoff as restructuring, cost discipline, or organizational simplification. The 2U/TWOUQ sensitivity check makes the same power problem visible in practice, since one distressed firm can move the treated-group estimate when strict AI disclosures are still rare, even though the result remains statistically insignificant either way.

Several limitations follow from this setup, beginning with the fact that eighteen strict AI events are too few to rule out small or uneven AI-framing effects and the estimates are too imprecise to support a tight zero. The strict coding also cannot fully separate real productivity information from AI framing, since both can appear in the same announcement. A firm may genuinely be replacing labor with AI, or it may be using AI language to make ordinary cost cutting sound more strategic, and an event study can only measure the market reaction to the disclosure. The hybrid CRSP62 and Yahoo Finance return data also introduce source heterogeneity, which the regressions handle with return-source fixed effects but cannot remove completely. These limits narrow the claim to the current public-firm disclosure sample, where strict AI-cited layoffs do not earn a statistically reliable premium over ordinary layoff announcements.

The main implication is that AI language may be less powerful when it is attached to layoffs than when it is attached to products, investment plans, or earnings growth. Investors may like AI as a productivity frame, but a workforce reduction still forces them to ask why the firm is cutting labor in the first place. The current data do not show a reliable AI-framing premium, while still leaving room for smaller effects, uneven effects, or effects that become easier to detect as more firms make explicit AI-layoff disclosures. The paper's contribution is to isolate this disclosure category early and show that, in its current public-firm form, AI-layoff framing does not produce the consistent market reward that the AI environment might lead one to expect.

## **7. Conclusion**

This paper asks whether investors react differently when a public firm announces layoffs and explicitly connects the workforce reduction to AI adoption. The comparison is deliberately narrow because every

observation in the sample is already a layoff announcement, which keeps the test focused on whether disclosed AI-layoff framing changes the price response to a labor cut. That distinction matters because the paper is measuring how markets respond to the public language around layoffs, while the larger question of how AI affects employment inside firms remains harder to observe directly.

The current public-firm sample gives little support to the idea that strict AI-cited layoffs receive a reliable market premium over ordinary layoff announcements. After excluding 2U/TWOUQ, whose return path is dominated by severe financial distress, the immediate-window coefficient sits close to zero, and the longer windows do not show the kind of fading or reversal that would support a stronger overreaction interpretation. Investors may still view AI as valuable productivity news in other settings, especially when it is tied to products, investment, or earnings growth, but in this setting the AI label does not consistently turn a layoff announcement into favorable news.

The layoff-size interaction leaves one useful direction for later work. The short-window interaction coefficients are positive, which fits the possibility that AI framing carries more weight when the workforce reduction is large enough to look economically material. The result remains weak and imprecise, especially because only 15 strict AI events have usable layoff-size data, so it cannot support a firm conclusion. It does suggest that future work should pay attention to the scale of the labor cut instead of treating all AI-cited layoffs as the same kind of announcement.

The small treated group is part of the empirical setting; after combining Layoffs.fyi and EDGAR, coding strict AI connections by hand, and matching events to returns, only 18 strict AI-cited layoffs remain in the final sample. That makes statistical power limited, but it also shows how rare direct AI-layoff disclosure still is in the public record. Firms may use AI internally without saying so directly, or they may describe workforce changes as restructuring, cost discipline, or organizational simplification. For that reason, the paper should be read as early evidence on public AI-layoff framing, with the larger labor-substitution question left for richer data.

What this paper shows is that AI language, at least when attached to layoffs, does not receive the kind of consistent market reward that the current AI environment might lead one to expect. The evidence is too early

and too small to rule out smaller or uneven effects, but it pushes against the prediction that firms can reliably improve the market reaction to layoffs simply by presenting them through AI language. In the current sample, investors seem to treat AI-cited layoffs with more caution than the surrounding AI narrative might suggest. Future research can test the same question more sharply if explicit AI-layoff disclosures become more common, but the early public-firm evidence points toward ambiguity instead of a reliable AI-framing premium.

## 8. Appendix

**Table 2**

| Group        | Events | Mean<br>layoff<br>size | Median<br>layoff size | Layoff size<br>available | Mean<br>CAR[-1,+1] | Mean<br>CAR[-1,+5] | Mean<br>CAR[-1,+10] | Mean<br>CAR[+3,+10] |
|--------------|--------|------------------------|-----------------------|--------------------------|--------------------|--------------------|---------------------|---------------------|
| Strict AI    | 18     | 2,339                  | 1,000                 | 83.33%                   | -1.45%             | 0.49%              | -0.63%              | 0.66%               |
| Ambiguous AI | 4      | 1,000                  | 1,000                 | 25.00%                   | -6.13%             | -7.24%             | -3.35%              | 2.93%               |
| Non-AI       | 144    | 981                    | 180                   | 100.00%                  | -1.11%             | -3.03%             | -3.47%              | -1.11%              |

**Notes:** This table reports descriptive statistics by announcement category. Strict AI refers to layoff announcements that explicitly connect the workforce reduction to AI adoption, automation, AI-driven restructuring, or replacement of labor with AI-related technology. Ambiguous AI refers to announcements with AI-related language that does not clearly identify AI as the cause of the layoff. CARs are cumulative abnormal returns over the listed event windows. Layoff size availability reports the share of events in each group with usable layoff-size data.

**Table 5**

| Window      | Full sample coefficient<br>with 2U | Full sample<br>p-value | Final sample coefficient<br>without 2U | Final sample<br>p-value |
|-------------|------------------------------------|------------------------|----------------------------------------|-------------------------|
| CAR[-1,+1]  | -0.0468                            | 0.358                  | -0.0021                                | 0.938                   |
| CAR[-1,+5]  | 0.054                              | 0.121                  | 0.0363                                 | 0.251                   |
| CAR[-1,+10] | 0.0197                             | 0.57                   | 0.0283                                 | 0.418                   |
| CAR[+3,+10] | -0.0061                            | 0.844                  | 0.0166                                 | 0.463                   |

**Notes:** This table compares the Strict AI coefficient with and without 2U/TWOUQ. 2U/TWOUQ is excluded from the main sample because its event-window returns are dominated by severe financial distress and bankruptcy-style trading rather than normal AI-layoff announcement information. The sensitivity check illustrates the fragility of inference in a setting where strict AI-layoff announcements remain rare.

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## 10. AI Disclosure Statement

I used AI tools as a research and coding assistant while completing this paper. The largest use was in the data workflow for the event study. Since there is no ready-made dataset of public firms explicitly tying layoffs to AI adoption, AI helped me plan parts of the workflow, write and debug Python scripts, process Layoffs.fyi entries, organize EDGAR filing evidence, match layoff announcements to public tickers, and prepare the final event-study files. AI also helped format regression outputs, check table consistency, and flag data issues such as missing return observations and the 2U/TWOUQ distressed outlier.

I manually reviewed and coded the treatment variable used in the paper. I separated strict AI-cited layoffs from ambiguous AI mentions and ordinary non-AI layoffs based on whether the announcement explicitly connected the workforce reduction to AI adoption, automation, or AI-driven restructuring. AI helped organize the coding process, while the classification criteria and final coding decisions were my responsibility.

I also used AI to troubleshoot the empirical workflow, including CRSP62 and Yahoo Finance return collection, cumulative abnormal return construction, regression scripts, table generation, and figure preparation. The regressions, tables, and figures reported in the paper are based on the final dataset described in the Data section, which contains 166 public-firm layoff events and 18 strict AI-cited layoffs.

For writing, I used AI to keep the interpretation as close to the evidence as possible. All empirical claims, citations, tables, and conclusions were reviewed by me before submission. The paper's final claim is my own assessment of the evidence. In the current public-firm sample, strict AI-cited layoffs do not show a statistically reliable AI-framing premium, and the result should be read as early evidence because explicit AI-layoff disclosures are still rare.